RAJESHPANDI K

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PROFESSIONAL SUMMARY:

Final-year Electronics and Communication Engineering student specializing in Embedded Systems and Automotive Electronics. Experienced in electric vehicle development, embedded programming, PCB design, and IoT-based systems. Proficient in STM32, Embedded C, and circuit simulation with hands-on experience in designing, testing, and troubleshooting electronic systems.

EDUCATION:

B.E. Electronics and Communication Engineering – Final Year (Knowledge Institute of Technology | CGPA - 8+ | Graduation year-2027).

TECHNICAL SKILLS:

PROGRAMMING: Embedded C, C, Java.
PROTOCOLS: UART, CAN, LIN, SPI, I2C.
MICROCONTROLLERS: STM32, NXP S32K, ESP32, Arduino.
TOOLS: STM32CubeIDE, S32 Design Studio, KiCad, LTspice, TSMaster, MATLAB, Proteus.
DOMAIN: Embedded System, Automotive Electronics, IoT, ECU Testing, Circuit debugging.

PROJECTS:

Electric Bike Development Project (4 Months)

- Designed and assembled a complete electric bike prototype.
- Configured and programmed the Votol EM100 controller to achieve approximately 2700 RPM.
- Performed system integration, troubleshooting, and performance optimization.

Technologies: Motor Control, Automotive Electronics, Embedded Systems.

Railway Track Monitoring System (2 Months)

- Developed a sensor-based railway track monitoring and fault detection system.
- Implemented alert mechanisms for fault identification.

Technologies: Embedded C, Sensors, Microcontrollers.

SAE Drone Project (3 Months)

- Analysing propulsion system design and component selection including BLDC motor, ESC.
- Performed battery sizing and system testing.
- Contributed to system integration and troubleshooting.

INTERNSHIP EXPERIENCE:

Epicure Robotics — Embedded Systems Intern [2months]

- Developed and tested **STM32-based embedded control systems** for robotic automation applications.
- Worked on **Embedded C firmware**, including control logic, system sequences, UART communication, and hardware-level debugging.
- Debugged firmware and hardware interactions using **breakpoints, serial monitoring, fault/status information, and test procedures**.

Technologies: STM32, Embedded C, Hardware Debugging, Motor Controllers, UART, SPI, MQTT, Python, Image Processing.

ACHIEVEMENTS:

- Participant – National Level Indian E-Bike Championship, Bhopal.
- Project selected for MSME Regional Level Competition.

AREA OF INTEREST:

Embedded Systems, Electric Vehicle (EV) Technology, IoT-based Embedded Design